SILENT PRIMES AND THE VARIANCE OF THE COLLISION COUNT

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Abstract. The gate width theorem gives an explicit formula for the bin-deranging set. This paper develops two consequences. First, for primitive-root primes at each fixed lag, the silent primes (those with zero autocorrelation) are the prime factors of an explicit arithmetic sequence, and only finitely many exist. Second, computation suggests the collision count $C(g)$ across non-deranging multipliers ($p > b+1$) has variance proportional to $Q = \lfloor (p-1)/b \rfloor$ (conjectured, not proved).

1. Introduction

This paper studies the variance of the collision count and introduces the Dirichlet series $L^G(s)$.

The gate width theorem [1] proved that the digit function $\delta(r) = \lfloor br/p \rfloor$ has exactly $b-1$ bin-deranging elements for any prime $p > b$, and identified them as the Möbius family $g = -u/(b-u) \pmod p$ for $u = 1, \ldots, b-1$. The present paper asks two questions. First: as p varies, which primes are silent at a given lag? Second: how does the collision count fluctuate across the non-deranging multipliers?

The first question leads to an explicit arithmetic characterization of the silent primes. The second reveals a square-root variance law that connects the fractional field to the equidistribution theory of sequences modulo p. The Dirichlet series framework follows classical methods [2, 3].

2. Silent Primes

Definition 1. A prime $p > b$ is *silent at lag* ℓ if $R_p(\ell) = 0$.

For a prime p where b is a primitive root, $R_p(\ell) = C(b^\ell \bmod p)$. By the gate width theorem, $C(g) = 0$ if and only if $g \equiv -u/(b-u) \pmod p$ for some $u \in \{1, \ldots, b-1\}$.

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Theorem 2. *For a primitive-root prime $p > b$ at fixed lag $\ell \geq 1$, p is silent at lag ℓ (i.e., $R_p(\ell) = 0$) if and only if p divides*

$$N(\ell, u) = b^{\ell+1} + u(1 - b^\ell)$$

for some $u \in \{1, \ldots, b-1\}$. Only finitely many primes are silent at any given lag.

Proof. The condition $b^\ell \equiv -u/(b-u) \pmod{p}$ gives $b^\ell(b-u) + u \equiv 0 \pmod{p}$, i.e., $p \mid N(\ell, u)$. Since $N(\ell, u) \neq 0$ for $\ell \geq 1$, only finitely many primes divide it. □

Proposition 3. *The $b-1$ integers $N(\ell, u)$ for $u = 1, \ldots, b-1$ form an arithmetic sequence with first term $b^\ell(b-1) + 1$ and common difference $-(b^\ell - 1)$.*

Proof. $N(\ell, u) = b^{\ell+1} - u(b^\ell - 1)$. The common difference is $-(b^\ell - 1)$. □

ℓ	silent primes ($p > 10$)	from $N(\ell, u)$
1	$11, 13, 19, 23, 37, 41, 73$	factors of $91, 82, \ldots, 19$
2	$13, 17, 19, 29, 37, 53, 101, \ldots$	factors of $901, 802, \ldots, 109$
3	$11, 13, 19, 31, 47, 79, 97, \ldots$	factors of $9001, \ldots, 1009$

TABLE 1. Silent primes at the first three lags in base 10.

3. The Collision Parameter and the Sawtooth Form

For a non-deranging $g \neq 1$, the gate width proof introduces the collision parameter $c = b(1-g)^{-1} \pmod{p}$ with $c \in \{b+1, \ldots, p-1\}$.

Write $((x)) = \{x\} - \frac{1}{2}$ for the sawtooth function.

Theorem 4 (Sawtooth form)**.** *With the sawtooth notation above,*

$$C(g) = Q + \sum_{m=1}^{Q} \operatorname{sgn}\left(\left(\left(\frac{mc}{p} \right) \right) - \left(\left(\frac{mb}{p} \right) \right) \right),$$

where $Q = \lfloor (p-1)/b \rfloor$ and $c = b(1-g)^{-1} \bmod p$ is the collision parameter.

Proof. The collision condition at residue r with parameter c reduces to a comparison of sawtooth values at $m = 1, \ldots, Q$. For each m: if $((mc/p)) > ((mb/p))$, both r and $p - r$ produce collisions (by the bilateral symmetry $r \mapsto p-r$, which preserves the comparison). So each positive comparison contributes 2 collisions. If $((mc/p)) < ((mb/p))$, neither r nor $p-r$ collides, contributing 0. Writing $\mathbf{1}_> = (1+\operatorname{sgn})/2$ and doubling: the total collision count is $2 \cdot \sum_{m=1}^{Q} \frac{1+\operatorname{sgn}}{2} = Q + \sum \operatorname{sgn}$. □

The sawtooth form expresses the collision count as a signed comparison of two sequences modulo p: the sequence $mc \bmod p$ (controlled by the collision parameter) and the reference sequence mb (controlled by the base). The collision count measures how often the first sequence "leads" the second.

The collision count depends on c through the wrapping behavior of the sequence $mc \bmod p$. The exact relationship between c and $C(g)$ is governed by the sawtooth comparison above.

4. The Variance Conjecture

The collision count $C(g)$ varies across the non-deranging multipliers. We now characterize its fluctuations.

Definition 5. For a prime $p > b$, the *collision variance* is

$$V(p) = \frac{1}{|S|} \sum_{g \in S} \big(C(g) - \overline{C}\big)^2,$$

where $S = \{g \in \mathbb{F}_p^\times : C(g) > 0,\ g \neq 1\}$ and $\overline{C}$ is the mean of C over S.

Conjecture 6 (Variance scaling)**.** *The collision variance satisfies*

$$V(p) = \lambda_b \cdot Q + O(1),$$

where $\lambda_b > 0$ is a constant depending only on the base b, and $Q = \lfloor (p-1)/b \rfloor$.

Equivalently, the standard deviation of $C(g)$ over the non-deranging multipliers is $O(\sqrt{Q}) = O(\sqrt{p/b})$.

This conjecture has been verified computationally for all primes $p \leq 10000$ in bases $b \in \{3, 6, 7, 10, 12\}$. The constant λ_b depends on b and converges slowly as $Q \to \infty$. In base 10: $\lambda_{10} \approx 0.89$.

p	Q	$V(p)$	V/Q	$\sigma/\sqrt{Q}$
97	9	6.00	0.667	0.817
193	19	13.97	0.735	0.857
499	49	40.45	0.826	0.909
997	99	84.35	0.852	0.923
1999	199	173.10	0.870	0.933
4999	499	440.55	0.883	0.940
9973	997	884.83	0.888	0.942

Table 2. Collision variance in base 10. The ratio V/Q converges to $\lambda_{10} \approx 0.89$, giving $\sigma/\sqrt{Q} \to 0.94$.

Remark 7. The square-root scaling $\sigma(C) \sim \sqrt{Q}$ is the characteristic fluctuation of equidistributed sequences modulo p. The sawtooth comparison in Theorem 4 sums Q terms, each ± 1, whose signs depend on the equidistribution of $mc \bmod p$ relative to mb. When the signs are approximately independent (as occurs for generic c), the central limit theorem gives variance proportional to Q. This scaling is observed computationally; a rigorous proof that the signs are sufficiently independent remains open.

The collision count $C(g)$ therefore fluctuates around its mean $\overline{C} \approx Q$ with amplitude $\sqrt{Q}$. This is the same square-root fluctuation pattern that governs the error term in the prime number theorem under the Riemann Hypothesis: $\pi(x) - \mathrm{li}(x) = O(\sqrt{x}\log x)$. Whether the fluctuations of $C(g)$ and the fluctuations of $\pi(x)$ are controlled by the same mechanism (the zeros of an L-function) is the central open question.

5. The Dirichlet Series

Definition 8. The *fractional field Dirichlet series* at lag ℓ is

$$L^G(s,\ell) = \sum_{p>b} \frac{R_p(\ell)}{p^s}.$$

Theorem 9. *$L^G(s,\ell)$ converges absolutely for* $\mathrm{Re}(s) > 2$.

Proof. The trivial bound $0 \le R_p(\ell) \le p-1$ gives $|R_p(\ell)/p^s| \le p^{1-\mathrm{Re}(s)}$. The sum $\sum_p p^{1-s}$ converges for $\mathrm{Re}(s) > 2$. □

At each lag ℓ, the series runs over primes $p > b$. For primitive-root primes, $R_p(\ell) = C(b^\ell \bmod p)$, connecting the autocorrelation to the global collision count. For non-primitive-root primes, this equality does not hold in general.

If the variance conjecture holds, the individual terms fluctuate by $O(\sqrt{p/b})$ around the mean. Whether these fluctuations have additional analytic structure (meromorphic continuation, connection to L-functions) is an open question.

6. The Möbius Family

The deranging set $\{-u/(b-u) : u = 1, \ldots, b-1\}$ is a family of $b-1$ rational numbers, independent of p. In base 10:

$$-\tfrac{1}{9}, \quad -\tfrac{1}{4}, \quad -\tfrac{3}{7}, \quad -\tfrac{2}{3}, \quad -1, \quad -\tfrac{3}{2}, \quad -\tfrac{7}{3}, \quad -4, \quad -9.$$

Proposition 10. *The Möbius family satisfies:*

(1) ***Inversion symmetry.*** *$-u/(b-u)$ and $-(b-u)/u$ are multiplicative inverses. The deranging set is closed under inversion.*

(2) ***Complement element.*** *For even b, $u = b/2$ gives $g \equiv -1 \pmod{p}$.*
(3) ***Complementary structure.*** *The fractions $u/(b-u)$ for $u = 1, \ldots, b-1$ are the ratios of complementary parts of b.*

The reduction of the Möbius family modulo varying primes governs both the silent prime structure (Section 2) and the collision variance (Section 4). At each lag ℓ, the silent primes are those where b^ℓ reduces to a member of the family. The collision variance measures how the non-family elements distribute their collision counts.

7. Remarks

Toward the fluctuation sum. The decomposition of $L^G(s, \ell)$ into mean and fluctuation parts sets up the central question for the next paper: does the fluctuation sum

$$\Phi_x(s, \ell) = \sum_{b+1<p\leq x} \frac{C(b^\ell \bmod p) - \overline{C}_p}{p^s}$$

have meromorphic continuation beyond its half-plane of convergence? Assuming the variance conjecture (Conjecture 6), the individual fluctuation terms would be $O(\sqrt{p/b})$, suggesting (but not proving) convergence for $\mathrm{Re}(s) > 3/2$.

If F admits continuation to $\mathrm{Re}(s) > 1$ with controlled growth, the zeros of F in the critical strip would govern the distribution of collision fluctuations across primes, in direct analogy to how the zeros of $\zeta(s)$ govern the distribution of primes.

The variance constant. The constant λ_b in Conjecture 6 depends on b and converges slowly with Q. Its exact value is an open problem. The sawtooth form (Theorem 4) suggests that λ_b is determined by the equidistribution properties of the sequence $mc \bmod p$ as c varies over $\{b+1, \ldots, p-1\}$, connecting the variance to the discrepancy theory of modular sequences. The logarithmic divergence of $\sum 1/p$ that underpins the Mertens-type growth law was established by Mertens [4]; see also [3].

References

[1] A. S. Petty, Bin derangements and the gate width theorem, research note, 2024.
[2] H. Davenport, *Multiplicative Number Theory*, 3rd ed., Springer, 2000.
[3] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.
[4] F. Mertens, Ein Beitrag zur analytischen Zahlentheorie, *J. Reine Angew. Math.* **78** (1874), 46–62.

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